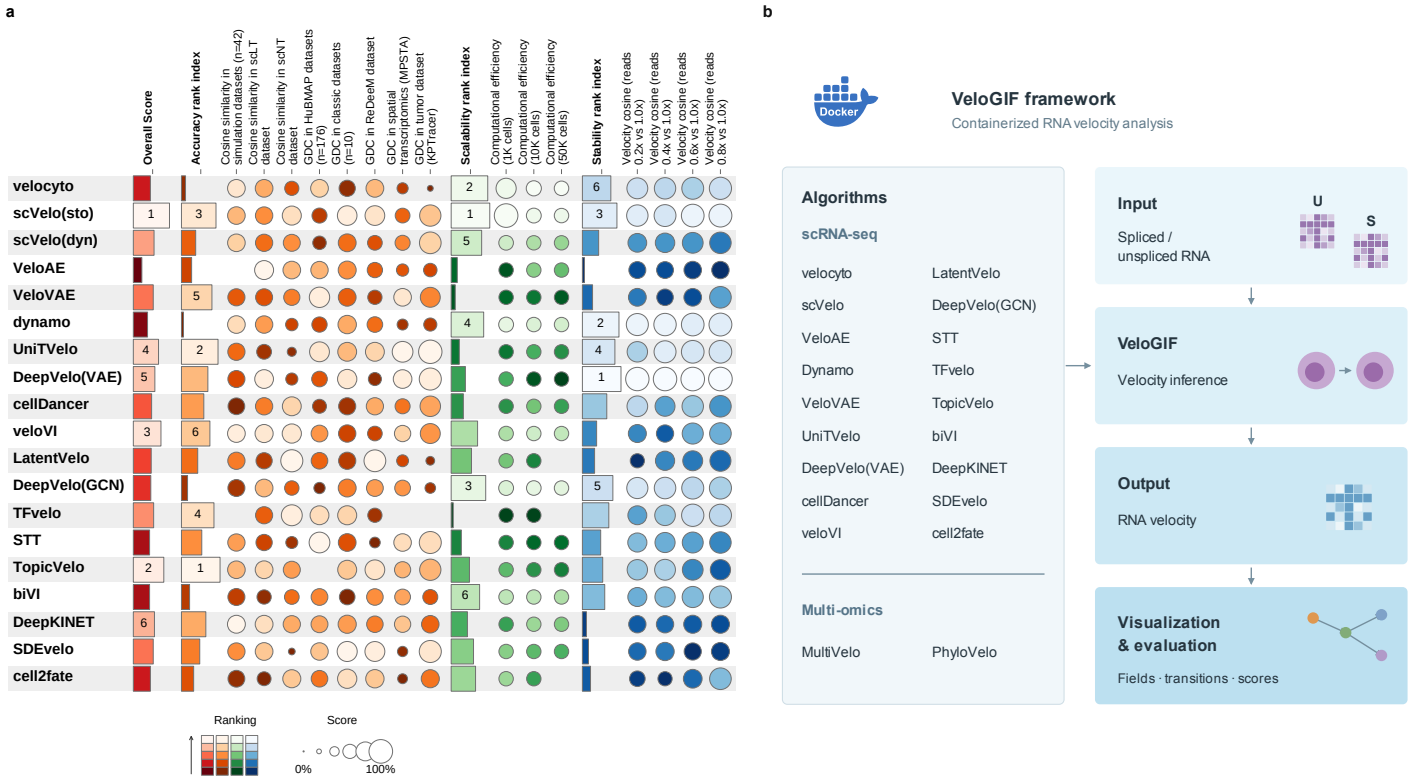


Fig. 6 | Overall performance with reads-downsampling stability and the VeloGIF framework. a, Summary of 19 method configurations across accuracy, scalability and stability. Stability uses the mean signed velocity cosine at 0.2, 0.4, 0.6 and 0.8× reads against 1.0× reads. The self-comparison at 1.0× is excluded. All 19 configurations have stability results. TopicVelo uses its integrated transition matrix projected into fixed full-depth common-gene Ms coordinates; its base scVelo velocity layer is not used. This transition-projected measure differs from the direct gene-velocity cosine used for cell2fate and other methods. Overall Score = 0.6×Accuracy rank index + 0.4×Scalability rank index; stability does not enter this score. Accuracy is aggregated over available domains among eight, and coverage/preprocessing differ among methods. Circle area maps signed cosine/GDC to (x + 1)/2 for visualization; signed values and coverage are retained in Source_data. **b,** Original VeloGIF framework artwork, with the method list extended to the current included methods. Method coverage varies across datasets.